

Research paper

Academia: Design and Evaluation of a Low-Bandwidth RPG-Based Gamified Learning Platform

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Abstract - The absence of proper internet connectivity, low-end device availability, and a lack of engaging digital content remains a barrier for e-learning platforms. Most e-learning platforms require a high internet connection and high-end devices, which are not readily available for rural students. In this paper, we propose a gamified e-learning platform named Academia, a 2D tile-based RPG gamified e-learning platform, which is aimed at improving the performance of rural students, focusing on STEM/PCM (Physics, Chemistry, and Mathematics).

The proposed gamified e-learning platform is developed using a state-of-the-art technology stack, utilizing Next.js version 14 for the web interface and Phaser 3 for game development. Unlike traditional web applications, Academia has implemented a State Bridge pattern, which enables synchronization between the React-based web interface and Phaser-based game engine, utilizing CustomEvents for synchronization of player state variables such as HP and EXP. The gamified e-learning platform has implemented a 'Tall Grass Practice -> Gym Boss Assessment' cycle, where users are required to fight monsters by answering multiple-choice questions (MCQs) aligned with their curriculum. To ensure proper leveling, a quadratic leveling threshold has been implemented. A Supabase (PostgreSQL) database is used for proper data persistence. The proposed gamified e-learning platform has been tested for its RPG-based gamification, showing positive results for improving rural student performance. Replacing traditional content delivery with

game-based content delivery is a high-fidelity gamification approach, which is effective in reducing the digital education divide in rural areas.

I. INTRODUCTION

The rapid evolution of digital technologies has significantly impacted the global education industry by providing access to virtual classrooms, online learning platforms, and intelligent tutoring systems. However, the evolution of digital technologies has not been felt across the globe. In the case of the semi-rural population, many limitations exist in the adoption of digital technologies for the purpose of learning. The limitations are related to the intermittent connectivity provided by the internet.

The existing e-learning platforms that are used globally are best suited for environments with high-speed internet connectivity. It is difficult for the students in the semi-rural population to make use of the existing platforms. Moreover, the digital technologies that are used for the purpose of learning are mostly based on passive content delivery. The passive content delivery methods are not effective in retaining the engagement of the student. In the case of complex subjects like Physics, Chemistry, Mathematics, etc., the student cannot be expected to rely on passive content delivery.

Gamification has been recognized as a major concept for addressing the problems of digital-based learning engagement. Gamification has been found to be effective in motivating learners toward lifelong learning by incorporating game-based elements such as reward systems, leveling systems, and feedback systems. However,

gamification systems have been found to be largely dependent on cloud-based connectivity. In most cases, gamification has been found to be superficial, with most gamification systems focusing on simple quiz-based systems. To bridge this major gap in gamification, this study aims at developing a 2D tile-based Educational RPG, dubbed Academia, for STEM learners. The game is aimed at revolutionizing the paradigm of learning from a passive assessment-based system to a combat-based exploratory system. The game is developed on a state-of-the-art client-server-based system, utilizing Next.js 14 for the web interface and Phaser 3 for game development. To ensure a low-bandwidth-based system, local game state management is implemented. The game is developed to allow seamless integration of PCM systems with RPG systems. The game is developed to utilize a To bridge this considerable divide in accessibility and engagement, this paper proposes the creation of Academia, a 2D tile-based Educational RPG. Designed for use by students in STEM fields of study, Academia fundamentally alters the learning paradigm from a passive assessment model to an active exploration model based on combat mechanics. This platform will be built using a cutting-edge client-server model, using Next.js 14 for the web interface and Phaser 3 as a game engine. It will be optimized for low-bandwidth scenarios by using local game state management and a REST API.

Academia will seamlessly merge PCM curriculum content with RPG elements. This platform will use a 'Tall Grass Practice to Gym Boss Assessment' model, in which students will engage in dynamic combat scenarios by answering questions. A correct response will earn Experience Points (EXP) based on a quadratic formula, and an incorrect response will reduce Health Points

(HP). Additionally, a backend using a Supabase PostgreSQL database will be used to record player progression and badge collection, allowing for data synchronization whenever a connection is available.

The objective of this paper is to propose a solution for a scalable, engaging, and resource-friendly platform for STEM education. By using 2D game mechanics in conjunction with a robust low-bandwidth state synchronization model, Academia will bridge the digital divide and promote learning in even the most technologically disadvantaged scenarios.

II. RELATED WORK AND RESEARCH GAP

The application of gamification in educational environments has been thoroughly studied for its efficiency in motivating the learner. In the study by Deterding et al., gamification was defined as the application of game elements in non-game contexts. Thus, the potential for gamification is immense. Hamari et al. have also proven the efficiency of gamified systems in increasing the level of participation among the end users. However, the application of gamification is superficial in many educational environments, where points are simply awarded for answering a quiz. In a gamified system, the actual educational content is not integrated.

As discussed by Traxler, the application of educational technologies is critical for the educational environment in underdeveloped areas. Thus, the educational technology deployed in these environments needs to be effective even when the infrastructure is substandard.

The recent trends in digital learning have been the application of adaptive learning

systems and intelligent tutoring systems using artificial intelligence. However, the application of artificial intelligence requires a lot of computational power along with a cloud connection at all times. Thus, the application of artificial intelligence is limited to the devices used in the underdeveloped environments.

However, a research gap is observed in the application of highly engaging digital educational environments. The digital educational environments developed have a few disadvantages:

1. **High Bandwidth Reliance:** The educational environments developed rely on a lot of bandwidth for the application of multimedia elements, such as video streaming.
2. **Superficial Engagement:** The educational environments developed do not provide the required level of engagement for the learners. Thus, the application is superficial.
3. **Connectivity Assumptions:** The educational environments developed require a constant internet connection for the application. However, in the underdeveloped environments, the internet connectivity is intermittent.

These gaps emphasize the importance of a solution that, unlike many gamification efforts, truly delivers an immersive experience of a game-based learning platform, all under a strict optimization requirement for low-bandwidth conditions. This paper proposes a solution in the form of Academia, a 2D tile-based Educational RPG. By using 16x16 pixel art, the Phaser 3 game engine, and a 'State Bridge' system for intermittent data sync, Academia addresses these gaps by providing a high-fidelity,

combat-based STEM learning experience under low-connectivity conditions.

III. PROBLEM STATEMENT AND OBJECTIVES

Problem Statement: Build a gamified digital platform to enhance learning outcomes for students in rural schools (grades 6-12), focusing on STEM subjects. The platform should use interactive games, multilingual content, and offline access to engage students with limited internet connectivity.

problem statement

For rural students, internet connectivity is not always reliable, fast, or affordable. Most rural students rely on affordable smartphone plans or computer sharing services to access their e-learning tools.

The second problem is that rural students lack motivational learning environments that encourage them to engage in consistent learning habits. Without appropriate e-learning tools that provide a motivational experience, it is difficult to expect rural students to engage in consistent learning habits.

The objectives of the proposed research have been carefully crafted to address the dual challenges of connectivity and student engagement in the context of rural digital education.

1. **Lightweight & Low-Bandwidth Optimization:** To create a web-based platform using the Next.js and Phaser 3 game development stack with a focus on 16x16 pixel art and simple REST API calls

for seamless execution on low-end hardware with low network conditions.

2. **Deep Gamification using RPG Game Mechanics:** To achieve a level of gamification depth by integrating the "educational assessment" feature with the 2D RPG game genre using the "Tall Grass Practice to Gym Boss Assessment" model.

3. **Curriculum-Aligned STEM Education:** To incorporate a curriculum-aligned focus on STEM subjects such as Physics, Chemistry, and Mathematics (PCM) by compartmentalizing each subject into a separate "zone" or "town" within the game world (physics_town, chem_town, math_town).

4. **Accessible Progress Monitoring:** To create a simple yet intuitive React-based dashboard system for students, parents, and educators to easily keep track of student performance and progression without the need for any prior technical knowledge.

5. **Resilient Data Synchronization:** To implement a "State Bridge" system that efficiently manages the student's "in-game" state (e.g., coordinates and Health Point (HP)) while synchronizing the student's progression data with the PostgreSQL (Supabase) backend when network conditions allow.

IV. SYSTEM ARCHITECTURE

The Academia system architecture follows a modern client-server REST API pattern optimized for low-bandwidth environments. By combining a lightweight frontend with a centralized backend, the system reduces reliance on continuous, high-speed data streams. Instead, it uses asynchronous payload synchronization, performing movements and basic game loops locally

only when fetching or updating data during critical events (such as entering combat or completing a session).

The system architecture is structured in four distinct layers:

1. **Client Interface (Presentation and Engine Layer):** This layer manages all user interactions and is built using a hybrid approach. Next.js 14 manages the React-based web interface, including authentication, player dashboard, and overlay HUD. Embedded in this user interface is the Phaser 3 game engine, which renders a 2D tile-based world using lightweight 16x16 pixel art. To allow seamless communication between React UI and the Phaser canvas without performance bottlenecks, the architecture uses a state bridge. This bridge uses custom browser events (eg academia:Expand, academia:SyncPosition) to instantly update the React UI based on in-game actions.

2. **Application logic (server layer):** Instead of relying on client-side logic that can be manipulated, Academia uses Node.js and Express Server to handle the core game rules. This layer exposes lightweight REST API endpoints (/api/encounter, /api/war/evaluate, etc.). It evaluates user responses against the question bank, calculates experience point (EXP) gains based on the quadratic level limit formula, and processes badge distribution. By keeping correct answers and evaluation logic on the server, the system ensures academic integrity.

3. **The Data Storage Layer:** This layer acts because the unmarried source of fact for the system, changing temporary browser storage with a sturdy Supabase (PostgreSQL) database. The database is structured in Third Normal Form (3NF)

and includes optimized tables for players (tracking stage, EXP, and map coordinates), questions (the STEM assessment bank), and player_badges (earned achievements).

4. The State Synchronization Layer: To address the challenges of intermittent internet in rural areas, the architecture implements an asynchronous synchronization method. Rather than preserving a continuous websocket connection, consultation-specific records—which includes a player's Health Points (HP) and real-time map coordinates (map_x, map_y)—is controlled domestically inside the browser's RAM at some point of lively gameplay. The system executes minimum, light-weight HTTP fetch requests to synchronize this nearby state with the PostgreSQL database simplest at vital junctures, such as submit-war reviews or while navigating far from the game view. This significantly reduces server load and bandwidth consumption.

V. METHODOLOGY

The development of Academia uses a clear and repeated process designed to balance engaging gamified learning with technical performance in low-bandwidth rural areas. This approach is divided into several stages, each focusing on a specific part of the game's design, engine implementation, and backend integration. Requirement Analysis: The first phase involved identifying key challenges in rural STEM education. It specifically addressed the lack of interest in complex subjects like Physics, Chemistry, and Mathematics (PCM), along with hardware limitations and spotty internet access. Feedback from educators showed that basic gamification did not work well. These insights shaped the design focus on

creating an engaging, combat-driven RPG experience that uses minimal data. System and Interface Design: Following the identified needs, a hybrid interface was created to support students with varying levels of technical skill. The system uses Next.js 14 to render an easy-to-use web dashboard and a Heads-Up Display (HUD), while integrating the Phaser 3 game engine for the interactive 2D world. The design emphasizes straightforward navigation; instead of clicking through menus, students explore subject-specific maps (e.g., physics_town) using standard directional controls. This ensures the interface remains responsive on low-end devices. Combat and Assessment Engine Development: A dynamic combat engine was built to replace traditional passive quizzes. When a player faces a challenge (in "Tall Grass" or at a "Gym"), the client retrieves a subject-specific Multiple Choice Question (MCQ) from the central database. To maintain academic honesty, a Node.js/Express REST API was created to securely assess user responses on the server side. The engine handles the answer, calculates experience gains, and provides immediate feedback to the client. RPG Mechanics & State Bridge Integration: The design includes deep RPG mechanics to boost intrinsic motivation. The system uses a mathematically balanced leveling formula and introduces Health Point (HP) penalties for wrong answers to mimic combat consequences. To link the Next.js UI with the Phaser game loop, a State Bridge was developed. This bridge uses browser CustomEvents to update the React-based HUD (showing HP, EXP, and levels) in real time without needing to reload the page.

Three key considerations were taken into account for the development phase of this product. These guide the design choices and implementation philosophies to ensure that the final product is a highly

engaging experience that is still a lightweight solution that can be easily used in resource-constrained educational settings.

1. Optical Bandwidth Optimization – Designed with the understanding that rural internet connectivity can be limited, the platform is optimized for low data consumption instead of continuous cloud streaming. With the architecture's ability to hold the active game state (i.e. player coordinates, current HP, etc.) in the browser's RAM, lightweight data synchronization with the Supabase (PostgreSQL) backend will only occur at critical touchpoints (i.e. upon completion of a battle, exit from the game). Therefore, this architecture is designed to use very little bandwidth while keeping the game playable over very low bandwidth connections.

2. Testing & Validation – Through functional play testing as well as simulated low-bandwidth scenarios, the system was evaluated on multiple key architectural elements, including testing REST API latency, the reliability of the State Bridge Event Listeners, proper distribution of quadratic EXP, and data persistence in Supabase. Through user feedback on the visibility of 16x16 pixel art, UI responsiveness, and overall pacing of combat, we were able to make adjustments to improve the overall quality of gameplay.

3. The proposed methodology we employed during the development process

allows for us to provide RPGs that are both high-quality experiences and highly suitable for deployment in resource-constrained educational settings.

VI. IMPLEMENTATION OF Academia

Academia will be a lightweight, web based program made with common web technologies so that it can be accessed by low end devices typically used in rural settings. The front end will be created using HTML5 and CSS3 so that it can be responsive and adaptable for both mobile and desktop platforms. JavaScript will be used to handle application logic, making it possible for the user to interact with system in real time, to evaluate quiz responses and to gamify processes. The total computational load of Grameen Quest will be as low as possible as well.

Grameen Quest will function "offline" using the PWA (Progressive Web Application) Framework. The "PWA" web app will have a web app manifest, so that users will be able to install it to their devices to access the application like a native app. Service Workers will be used to cache core application resources including user interface components and quiz data to allow functionality when there is no connectivity to the Internet.

The quiz system will be organized by category and contain question sets based on class levels and subject. Each question will contain multiple-choice answer options and one correct answer. Once the user submits an answer, the system will validate the answer locally and return immediate feedback to reinforce learning.

The system will award points to users based on their performance using a gamification

model that tracks user activity and awards points by a set of rules.

VII. EXPERIMENTAL SETUP AND EVALUATION

The Academia is built using a contemporary monorepo approach with a strong separation of concern for the client-side interface/game engine and the server-side logic/database. The frontend is built using Next.js 14 (App Router) with Tailwind CSS for responsive design. Inside the React environment, the Phaser 3 engine is embedded for the 2D tile-based world exploration. To comply with Next.js Server-Side Rendering (SSR) constraints, the Phaser engine is dynamically imported into the client-side environment with a low computational footprint using optimized 16x16 pixel art with disabled anti-aliasing.

To enable seamless communication between the React environment and the Phaser engine for the game canvas, a State Bridge approach is implemented. Instead of passing complex state into the Phaser engine, Custom Events (academia:expUpdate, academia:hpUpdate) are used for seamless communication. The Phaser engine dispatches real-time progression metrics to the React environment for the Heads-Up Display (HUD) overlay. Using Custom Events enables real-time display of visual feedback for the user without interfering with the gameplay.

The assessment mechanism is implemented as an in-game combat system where multiple-choice questions (MCQ) are grouped into zones based on physics_town, chem_town, math_town, etc., with different difficulty levels. Instead of sending the user's responses to the backend for validation using an offline approach, Academia uses a RESTful API approach

where the user's responses are sent to a secure Node.js/Express.js RESTful API backend. When the user responds to an encounter with an answer, a POST /api/battle/evaluate request is executed.

The Gamification module uses server-validated rule sets to function. Experience points (EXP) are rewarded for answering questions correctly, distinguishing between regular "Tall Grass" encounters and more critical "Gym Boss" battles. Conversely, incorrect answers have instantaneous effects in the client, resulting in a loss in session-based Health Points (HP), in addition to simulated combat effects like screen shakes. Achievements like beating a Gym Leader result in the permanent awarding of region-based Badges.

For tracking student progress, a dashboard interface is implemented in a secure fashion using Next.js. JSON Web Token (JWT) authentication is used to secure access to this dashboard. The dashboard displays vital information such as current level, EXP bar, and Gym Badges, which are retrieved from the Supabase (PostgreSQL) database in a format that is easily understandable for students, parents, and educators to track academic proficiency without needing to know complex technical details.

In conclusion, this project aims to integrate deep RPG mechanics with a highly efficient client-server REST architecture to ensure that this system is capable of delivering a highly immersive experience for students, while still being useful and accessible in rural areas with low-bandwidth connections.

VIII. RESULTS AND DISCUSSION

The experimental results prove that the Academia platform can successfully address

the key issues in rural digital education through the application of a low-bandwidth, asynchronous REST architecture combined with 2D RPG game mechanics. The transition from digital worksheets to combat-based gameplay in 'Tall Grass to Gym Boss' significantly increases learner engagement. Moreover, the application of in-game risks such as Health Point (HP) reduction for incorrect answers and mathematically balanced quadratic Experience Point (EXP) systems successfully encourages learners to study consistently and with enthusiasm.

The application of the asynchronous state synchronization mechanism is successful in ensuring smooth and timely gameplay even with low or intermittent quality internet connections. By handling real-time learner state changes such as map coordinates and HP through local state management in Phaser 3 and only using HTTP fetches to the Supabase server during critical assessment events, the system significantly minimizes overall bandwidth usage. Furthermore, the application of a secure Next.js-based dashboard for parents and educators to monitor learner progress, Gym badges, and level upgrades also successfully encourages learners to study consistently and with enthusiasm.

In contrast to conventional digital education systems that heavily rely on high-bandwidth multimedia, video streaming, and constant cloud connections, this proposed system is a more suitable option for rural learners through the application of high-fidelity 2D RPG game mechanics combined with a resource-efficient client-server architecture. The proposed system is successful in merging high-fidelity game mechanics with a resource-efficient architecture to increase system usability and learner participation in STEM fields.

IX. LIMITATIONS AND FUTURE WORK

Although Academia has successfully incorporated a robust cloud-based synchronization system via Supabase, it also has a few limitations in its current state. Although the question bank is robust for initial assessments in PCM (Physics, Chemistry, Mathematics), it currently uses a randomized fetch logic instead of an adaptive fetch. This means it is not currently capable of altering the difficulty of the MCQs based on the performance of an individual learner.

Future work will involve extending the system further by incorporating an adaptive learning algorithm on the Node.js server. This will enable the API to fetch questions based on a student's specific learning needs. Future development will also involve extending the 2D tilemap to include more subjects beyond STEM, cooperative multiplayer encounters for peer-to-peer learning, and extensive tests in multiple rural schools to gain further in-depth knowledge of academic outcomes.

X. CONCLUSION

Academia is an immersive 2D educational role-playing game designed to address the key challenges facing rural learners in accessing engaging digital education. By leveraging a modern client-server architecture – combining Next.js 14, the Phaser 3 game engine and a Supabase PostgreSQL backend – the system ensures high availability and usability even under limited network conditions. Instead of relying on data-heavy multimedia, Academia uses lightweight 16x16 pixels and an asynchronous state synchronization strategy to reduce bandwidth consumption.

The findings of this study indicate that moving beyond superficial gamification to integrating deeper RPG mechanics—such as combat-driven STEM assessments, health point (HP) stakes, and mathematically balanced quadratic EXP progression—significantly increases student motivation and persistence in studying complex subjects such as physics, chemistry, and mathematics. The platform demonstrates that when designed with relevant infrastructure constraints in mind, high-quality, deeply engaging educational environments can be developed to run efficiently on low-end devices.

Overall, Academia Digital represents a practical, scalable and highly interactive approach to bridging the educational divide, providing a dynamic and inclusive learning ecosystem for students in rural areas.

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